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Lab Number: 01

Problem #1:

1. take the first value from the list of values and assume it is the largest and smallest value
   Explanation: This step gives a starting point for the algorithm, by initializing and storing a value in the largest and smallest value.

2. while all the values in the list have not been compared:

   take the next value from the list of values, and make it the new value

   if the new value is greater than the largest value

   3.1 make the new value the largest value

   else

   if the new value is smaller than the smallest value

   3.1.1 make the new value the smallest value

   else

   if the new value and the largest/smallest value are equal, assume they are one value
   else
   put the new value to the back of the list

Explanation: This step will make sure each value in the list is compared, and would allow for the largest and smallest values to be determined.

3. Return the largest value and smallest value
   Explanation: This step will return the largest and smallest value.

Show by Hand:

List of Values: 8.7, 2, -9.8, -9.8, 10.0, 4.0, 17.0

1. Largest Value = 8.7
   Smallest Value = 8.7
2. New Value = 2
3. Smallest Value = 2
   New Value = -9.8
4. Smallest Value = -9.8

New value = -9.8
5. New Value = 10.0
6. Largest Value = 10.0
   New Value = 4.0
7. New Value = 17.0
8. Largest Value = 17.0
9. Largest Value = 17.0
   Smallest Value = -9.8

How the Algorithm will Ultimately Stop:
I have ensured that my algorithm eventually stops by adding a while statement in step two. This would allow for step two to be repeated until all the values in the list have been compared. Furthermore, step two would help with determining the largest and smallest values in the list of values, which would later be returned using the return statement, which also further ensures that the algorithm will ultimately come to an end.

Problem #2:

1. take one of the shortest sides and let it be represented by “a”, take the other shortest side and let it be represented by “b”
   Explanation: Initializes the shorter side lengths.

2. let the length of the hypotenuse be represented by “c”
   Explanation: Initializes the hypotenuse.

3. let “c” be equal to the sum of the values of “a” squared and “b” squared, and square root that sum.
   Explanation: Stores a value for c.

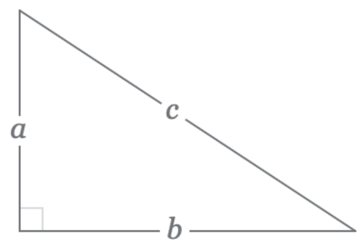
4. to calculate the parameter, let “parameter” equal to the sum of the values of “a”, “b”, and “c”
   Explanation: Initializes the parameter of the triangle and stores a value for the parameter.

5. let “area” represent the surface area of the triangle. To calculate the surface area of the triangle, let “area” equal the product of ½, “a”, and “b”
   Explanation: Initializes the surface area of the triangle and stores a value for it.

6. return c, parameter, and area
   Explanation: Returns the length of the hypotenuse, parameter of the triangle, and surface area of the triangle.

Show by Hand:

Let a = 3 and b = 4



1. a = 3
   b = 4
2. Hypotenuse = c
3. c = $\sqrt{3^2 + 4^2} = 5$
4. parameter = 3 + 4 + 5 = 12
5. area = $\frac{1}{2} \times 3 \times 4 = 6$
6. c = 5
   parameter = 12
   area = 6

How the Algorithm will Ultimately Stop:
I have ensured that my algorithm eventually stops by adding a return statement that would return the values which were calculated using the algorithm.

Problem #3

Algorithm:
1. Display the statement: “This is my first C program.”

Code:
```
/* My first C program */
#include <stdio.h>
int
main (void)
{
     printf ("This is my first C program.\n"); //print the statement
```

```
    return (0);
}
```

Output:

```
This is my first C program.
```

Screenshot:

```
This is my first C program.


=== Code Execution Successful ===
```